

PRATYUSH PRAKHAR

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PROFESSIONAL SUMMARY

Cybersecurity Professional with over 4 years of experience specializing in securing cloud-native applications and mitigating adversarial threats in AI/ML systems. Proven expertise in conducting whitebox, greybox, and blackbox security assessments, developing SAST/DAST pipelines, and implementing secure coding practices across CI/CD workflows using Python, Rust, and Golang. Experienced in addressing OWASP LLM Top 10 risks, adversarial ML techniques, and model integrity validation for Generative AI frameworks like LangChain and Amazon Bedrock. Adept at leveraging STRIDE threat modeling, MITRE ATT&CK framework, and zero-trust architecture principles to reduce attack surfaces by up to 30%. Strong collaborator with a track record of enhancing organizational security posture through automation, innovative tooling, and cross-functional engagement.

TECHNICAL SKILLS

- **Cloud & Application Security:** AWS IAM/S3/EC2 Hardening, STRIDE Threat Modeling, Application Security Design Reviews, SAST/DAST Pipeline Design, PCI-DSS/NIST Compliance, SIEM/SOAR Implementation, Zero-Trust Architecture Principles.
- **GenAI Security Specialization:** OWASP LLM Top 10 Mitigation, Adversarial ML Defense (Data Poisoning, Model Inversion), AI Supply Chain Security, LangGraph/AutoGen/CrewAI Security Frameworks, AWS Bedrock.
- **Offensive Security and Tools:** Penetration Testing (OSCP Methodology), MITRE ATT&CK Mapping, RPL/SMB Exploit Development, Burp Suite, Metasploit, Wireshark, Nmap, Nessus, sqlmap, hydra.
- **Programming Languages:** Golang, Python, Rust, Bash/Shell Scripting, JS, SQL, C/C++, Bash.
- **AWS Tools:** AWS Console, AWS Toolkits, AWS Developer Tools, AWS RDS.
- **GenAI Frameworks and Tools:** OpenAI, Llama, Gemini, Anthropic, CrewAI, AutoGen, LangGraph.
- **Networking & Protocols:** TCP/IP, SSL/TLS, DNSSEC, HTTPs, RPL, SMB, SSH, OAuth 2.0, RCP.
- **Databases:** MongoDB, MySQL, DynamoDB

CERTIFICATIONS

- **Offensive Security Certified Professional (OSCP)** | In Progress - Expected December 2025
- **AWS Certified Security** | In Progress
- **PortSwigger Certifications:** OWASP Top 10, OWASP LLM Top 10 | Completed
- **TryHackMe Certifications:** Web Pentesting, Burp Suite, Metasploit | Completed

PROFESSIONAL EXPERIENCE

Sr. Security Analyst | Solgenie Technologies Inc, USA

January 2024 - December 2024

- Conducted comprehensive risk assessments for digital applications, identifying critical vulnerabilities and formulating structured remediation plans based on risk indices. Reduced cross-AWS application communication attack vectors by 20% through targeted security controls.
- Collaborated with cross-functional teams to elevate security standards by implementing robust code reviews, vulnerability testing, and knowledge-sharing sessions on industry-standard tools such as Burp Suite and Metasploit.
- Delivered consultative security assurance services to clients, effectively communicating vulnerability impacts and actionable remediation strategies. Reduced security review assessment time by 40%, saving \$10K monthly in operational costs.
- Leveraged Generative AI to automate incident response processes, streamlining workflows and improving assessment accuracy by 10%. Implemented OWASP Top 10 compliance measures, focusing on API and microservice security.

Application Security Engineer | Amazon Web Services, New York, USA

July 2021 - December 2023

- Conducted comprehensive security design reviews for high-risk AWS services (e.g., AWS CodeSuite, IAM Console, SDKs), ensuring adherence to global security best practices. Identified and mitigated vulnerabilities through blackbox testing, reducing breach risks across 15+ services.
- Directed security initiatives for Amazon Bedrock's AI services, implementing guardrails against adversarial ML attacks. Reduced threat exposure by 25% within four assessment cycles by integrating OWASP LLM Top 10 principles.
- Performed whitebox, greybox, and blackbox security assessments to uncover architectural flaws. Spearheaded static code analysis using SAST tools, detecting and remediating 150+ critical vulnerabilities early in the SDLC.
- Applied STRIDE and MITRE ATT&CK frameworks to advanced threat hunts, preemptively mitigating risks in cloud environments. Enhanced service security posture by 10-15% through targeted vulnerability remediation efforts.
- Designed and integrated Python-based SIEM solutions and OS patching workflows into CI/CD pipelines, improving incident response efficiency by 30% and minimizing manual intervention.
- Authored service-level documentation to elevate organizational security awareness. Collaborated with product teams to integrate secure SDLC practices, improving overall security posture across multiple AWS services.

Application Security Analyst | Varian Medical Systems, Palo Alto, USA

June 2020 - September 2020

- Designed and implemented critical security administration frameworks within the SAP landscape, safeguarding high-risk applications such as HR-Workday. Enhanced access security by deploying role-based access controls (RBAC).
- Spearheaded enhancements to SAP role-based access management systems, ensuring compliance with industry standards

and reducing unauthorized access incidents by 20%.

- Refined audit controls for the SAP S/4HANA suite, optimizing PFCG business roles within the CRM system to bolster security for a workforce of over 10,000 employees. Reduced access vulnerabilities by 15% through meticulous RBAC.
- Engineered an innovative prototype automating ServiceNow ticketing workflows, eliminating redundancy and increasing team efficiency by 6 hours weekly. Achieved seamless integration with SAP systems to streamline issue management.

Security Analyst | SPRITZ Security and Privacy Group, Italy

June 2018 - September 2018

- Designed and implemented an Intrusion Detection System (IDS) for Smart Grid architectures to counter Sybil and Wormhole attacks in RPL-based Industrial IoT networks. Achieved a 98% detection accuracy by leveraging the Highest Rank Common Ancestor verification principle through rigorous emulation.
- Developed a proprietary wormhole detection methodology, integrating advanced network security concepts into CONTIKI-OS-based COOJA Simulator. Coded the network architecture in C/C++, while employing Java for simulation scenario construction.
- Conducted comprehensive testing to fortify IoT network security frameworks against advanced attack vectors, ensuring robust protection for critical infrastructure systems.

EDUCATION

M.S., Project Management - Cyber Security Operations, Harrisburg University

August 2024 - July 2025

Coursework – Agile/Scrum Methodology, Securing App and Software, Cloud and Network Security

GPA: 3.85/4.00

Master of Science in Cyber Security Engineering, University of Southern California

August 2019 - May 2021

Coursework - Web Application Security, Security Assurance & Privacy, Cryptography

GPA: 3.85/4.00

Bachelor of Technology in Electronics and Communication, Delhi Technological University

August 2015 - May 2019

Coursework - Computer Networks, Operating Systems, Algorithms & Data Structures

GPA: 8.22/10.00

RESEARCH AND DEVELOPMENT CONTRIBUTIONS

GenAI Security Contribution

Ongoing

- Designing end-to-end security pipelines to streamline threat detection and response using novel Generative AI solutions, including LangGraph and AutoGen frameworks.
- Enabling streamlined developer contributions to secure AI-driven systems by automating threat detection and response workflows in upcoming GenAI development applications.
- Engineering a cutting-edge **Capture the Flag (CTF) solver, OneForAll**, automating exploitation workflows by integrating tools like Nmap, Metasploit, SQLMap, and Hydra. This tool will leverage advanced GenAI capabilities to address complex cybersecurity challenges, including prompt injection attacks and adversarial ML techniques.

Open-Source Contribution

Ongoing

- Authored detailed documentation to simplify the learning curve for newcomers to the fields of Generative AI and cybersecurity, bridging the gap between theoretical concepts and practical applications.
- Published detailed writeups for well-known CTFs and Machines on GitHub to guide newcomers in field of pentesting and cybersecurity.